

Project Proposal

Project Title: Multi-Objective Reinforcement Learning for Robust and Energy-Efficient Quadrupedal Locomotion

Team Members:

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1. Problem Statement

Current control strategies for quadrupedal robots often prioritize velocity tracking and terrain traversal at the expense of energy efficiency. While Deep Reinforcement Learning (DRL) has demonstrated exceptional capability in generating dynamic gaits, standard reward-shaping techniques frequently result in high-torque, high-impact behaviors that are mechanically stressful and energetically wasteful. This project addresses this limitation by implementing a constrained optimization framework to enforce strict energy budgets while maintaining robust locomotion under varying environmental conditions.

2. Objectives & Scope

The primary objective is to develop and benchmark a locomotion policy for the **Unitree A1** quadruped that optimizes the trade-off between agility and power consumption.

- **Core Objective:** Implementation of **Proximal Policy Optimization (PPO)** to achieve stable velocity tracking.
- **Research Hypothesis:** A Constrained Markov Decision Process (CMDP) formulation (treating energy as a hard constraint) yields a superior performance profile compared to traditional scalarized reward shaping.
- **Advanced Objective (Robustness):** Evaluation and fine-tuning of the efficient policy against environmental perturbations, specifically variable ground friction and external force application.

3. Methodology

The project will be executed using a three-phase development lifecycle:

- **Phase I: Baseline Development.** Deployment of the Unitree A1 model within the **MuJoCo** physics engine. Training of a baseline "Teacher Policy" focused solely on velocity tracking and gait stability to establish performance benchmarks.
- **Phase II: Constrained Optimization.** Integration of energy metrics (Mechanical Power and Control Effort $\Sigma \tau^2$) into the learning process. Comparison of soft-constraint (weighted sum) vs. hard-constraint (Lagrangian) optimization methods to minimize Cost of Transport (CoT).
- **Phase III: Robustness & Analysis.** Subjecting the optimized agents to domain randomization (friction variability, mass uncertainty, and push perturbations) to ensure the efficient gait remains viable in non-ideal conditions.

4. Tools & Technologies

- **Simulation Platform:** MuJoCo (Multi-Joint dynamics with Contact).
- **Robot Model:** Unitree A1 (URDF/MJCF).
- **Machine Learning:** PyTorch, Gymnasium.
- **Algorithm:** PPO (Proximal Policy Optimization).

5. Expected Deliverables

1. A high-fidelity simulation environment for the Unitree A1 in MuJoCo.
2. A trained RL agent capable of efficient locomotion with quantifiable energy savings over a baseline model.
3. A technical report benchmarking the stability and energy metrics of different optimization strategies.
4. Visual demonstration of the agent's recovery capabilities under external perturbation.